

Kashmir Valley Transit

Headway & Fleet-Sizing Fundamentals

A study guide to the maths and reasoning behind the v3.3.5 Kashmir Valley Transit Rationalisation Engine — covering how headway is chosen, how cycle time is estimated, how fleet size is derived, and how every number on the dashboard ties back to a formula or a CHALO observation.

Engine v3.3.5 • May 2026

Principal Secretary of Transport, J&K

1. Why headway is the most important number

Headway is the time gap between two consecutive buses on the same route. Everything else in a transit plan — fleet size, service hours, vehicle type, peak-vs-off-peak operations, revenue, even bus-stop infrastructure — is downstream of this one decision.

A passenger picking between bus, auto, or private vehicle is really picking between **wait time + travel time**. Wait time is approximately $\text{headway} / 2$ on average (you arrive at a random moment, the next bus is on average half a headway away). So a 20-minute headway means a 10-minute average wait, which most people will tolerate; a 60-minute headway means 30 minutes of waiting, which most won't.

Average passenger wait time $\approx$ Headway / 2 (uniform arrival assumption)

This is why the engine treats headway as a **policy lever**, not a derived value. Every active route is assigned a target headway based on its priority band; everything else follows.

Headway targets in the Kashmir Valley v3.3.5 plan

Band	Description	Headway	# of routes
SSCL	30 CHALO e-bus routes (45 matched permits)	15 min	45
HP	High-priority non-SSCL trunks	20 min	85
MP	Medium-priority feeders	35 min	50
LP	Low-priority / lifeline routes	60 min	23
Social-promoted	Social-flag LP→MP exceptions	45 min	4

Why these particular numbers? Indian peer-city benchmarks (see Appendix A) show urban trunks at 5–15 minutes in the biggest metros, 15–25 minutes in tier-2 cities, and 25–35 minutes in tier-3 cities. Srinagar's size (1.66M residents) puts it between Chandigarh and Bhopal — so 15–20 minutes on the busiest corridors is the achievable phase-1 target.

2. Cycle time — what one bus actually takes per round trip

Cycle time is the wall-clock time for one bus to complete one round trip on a route: drive from origin to destination, (short break/layover), drive back. Once you know cycle time and headway, the fleet count follows mechanically.

```
Cycle_Time_Min = drive_time + stop_dwell + junction_penalty +  
bridge_penalty + layover
```

Each ingredient, in the Kashmir engine

- **Drive time** — OSRM routing gives the free-flow travel time between origin and destination. The engine fetches this from a local Docker OSRM instance with a Kashmir/India OSM extract. Multiplied by a congestion factor: **2.2× in Downtown Srinagar** (latitude > 34.07°), **1.4× peri-urban, 1.0× rural**.
- **Stop dwell** — every bus stop costs ~30 seconds for boarding/alighting. The engine estimates N stops as 1 stop per 500 m, then adds $STOP_PENALTY_MIN \times N_stops$ to the cycle. $STOP_PENALTY_MIN = 0.5 \text{ min}$.
- **Junction penalty** — major intersections, signals, and traffic circles each add ~30 s. Detected from sharp turns in the OSRM geometry.
- **Bridge penalty** — Jhelum river crossings add a flat **JHELUM_BRIDGE_BOTTLENECK_MIN = 8 minutes** in both directions, captures the Habba Kadal / Amira Kadal peak-hour queuing that OSRM's free-flow duration cannot see.
- **Layover** — buses need a short break at terminals. Engine adds ~10% of drive time, capped. Industry standard.

Worked example — Batamaloo ↔ Hazratbal (SSCL-03)

Component	Value	Notes
Route_KM (round trip)	20 km	OSRM measured
Drive @ 14 km/h avg	85.7 min	2.2× downtown congestion
Stop dwell (40 stops × 0.5 min)	20 min	1 stop per 500 m
Junction penalty	4 min	8 major junctions × 30 s
Jhelum bridge	8 min	Constant addition
Layover	8 min	10% of drive time
Cycle_Time_Min	~125 min	what the engine computes

Reality check: CHALO's empirical effective cycle on this route is closer to ~55 min, because buses skip low-demand stops at peak and use faster road segments. So the engine's 125 min is conservative — it assumes every scheduled stop is served and downtown congestion applies at peak. That conservatism is why the recommended fleet is higher than current operations.

3. Fleet sizing — the master formula

Once you have cycle time and headway, the number of buses needed to deliver service at that headway is mechanical:

$$\text{Fleet_Required} = \lceil \text{Cycle_Time_Min} \div \text{Headway_Min} \times \text{spare_ratio} \rceil$$

The ceiling function is essential — you cannot run half a bus. The **spare ratio** (engine default 1.15) accounts for scheduled maintenance, breakdown rotation, depot storage, and driver shift changes. Industry norm is 10–20%; the engine picks the middle.

Intuition — why the formula works

Picture a corridor with buses spaced exactly Headway minutes apart, all moving. Every Cycle_Time_Min the first bus arrives back at the start; in that span it has been overtaken (in time, not space) by exactly Cycle_Time_Min / Headway_Min other buses. That ratio IS the operating fleet you need on the corridor simultaneously.

Add the 15% spare and round up, and you have the fleet that delivers reliable service at the target headway. Drop any of the spare buses and headways start to slip — typically the bus going for service today means tomorrow's peak runs at Cycle_Time / (Fleet – 1), which is a longer headway than promised.

Worked examples from the v3.3.5 output

Route	Cycle (min)	Headway	Formula	Fleet
Batamaloo ↔ Hazratbal (SSCL-03)	125	15	$\lceil 125/15 \times 1.15 \rceil$	10
Lalbazar ↔ LD	47	20	$\lceil 47/20 \times 1.15 \rceil$	3
Parimpora ↔ Harwan (SSCL-01)	108	15	$\lceil 108/15 \times 1.15 \rceil$	9
TRC ↔ Pulwama (SSCL-15)	165	15	$\lceil 165/15 \times 1.15 \rceil$	13
Soura ↔ Ganderbal (LP)	75	60	$\lceil 75/60 \times 1.15 \rceil$	2

Two important nuances in the Kashmir engine

- **Floors** — urban/peri-urban routes get a minimum of 2 buses regardless of demand (covers the spare-rotation gap). Regional lifeline routes get a floor of 1.
- **SSCL empirical floor** — for the 45 SSCL-matched permits, the engine takes **max(formula_fleet, CHALO_empirical_fleet)**. If CHALO currently runs 5 buses on a route but the formula says the 15-min target needs 8, the engine recommends 8. If the formula says 3 but CHALO runs 5, the recommendation stays at 5 — never below current operation.

4. Capacity vs demand — Phase-4 KPIs

After fleet sizing is locked, the engine computes Phase-4 derived KPIs to flag routes where the recommended service is either too thin or too generous. These do not feed back into the fleet calculation — they are diagnostic outputs only.

Daily capacity (corrected in v3.3.3)

```
Daily_Capacity_Pax = Fleet × VehicleCapacity × (ServiceMinutes ÷ Cycle_Time_Min)
```

Service hours are 6:00 AM – 10:00 PM (16 hours = 960 minutes). Vehicle capacities: **50 seats HPV** (12-metre bus), **35 seats MPV** (9-metre bus), **15 seats LPV** (tempo). Each bus completes $\text{ServiceMinutes} / \text{Cycle_Time_Min}$ round trips per day; multiply by per-bus capacity for total seats offered.

Worked example — LALBAZAR ↔ LD

Quantity	Value
Fleet (9-metre MPV)	5 buses
Per-bus capacity	35 seats
Cycle_Time_Min	47.2 min
Service minutes / day	960 min
Round trips per bus per day	$960 / 47.2 = 20.34$
Daily_Capacity_Pax	$5 \times 35 \times 20.34 \approx 3,559$

Daily demand — the corridor-share model (v3.3.2+)

Demand on a route is harder than capacity, because residents in the 400-metre walking buffer might use this route, a parallel route, an auto, or a private vehicle. The engine apportions buffer population among overlapping routes using a **headway-weighted share**:

```
Daily_Demand_Pax = Pop_Buffer × corridor_share × mode_share × trip_rate × CAPTURE_SCALE  
corridor_share = (1 / Headway_Min × mean_inv_headway) ÷ (competitors × overlap_metric)  
mode_share = 0.09 urban / 0.072 peri-urban / 0.054 regional (typology-aware, CHALO-derived)  
trip_rate = 1.6 trips per resident per day  
CAPTURE_SCALE = 0.18 (empirically anchored to CHALO)
```

CAPTURE_SCALE = 0.18 is the only "calibration knob" — it ties the buffer-based supply model to CHALO's observed SSCL ridership (~32k/day across 30 routes). It absorbs auto/walk/private-mode leakage that the 400 m buffer otherwise over-counts. Re-calibrate if CHALO totals shift by more than ±15%.

Load_Flag — the diagnostic signal

Load_Ratio = Demand/Capacity	Load_Flag	Meaning
≤ 0	Red_NoCapacity	Fleet is 0 — service deactivated
$0 < r < 0.4$	Amber_Under	Capacity exceeds demand by 2.5×+ — could trim headway
$0.4 \leq r \leq 0.85$	Green	Sweet spot — load comfortable, room for surge
$0.85 < r \leq 1.0$	Amber_Tight	At capacity — peak crowding likely
$r > 1.0$	Red_Overload	Demand exceeds offered seats — add buses or accept crowding

v3.3.5 distribution: 9 Green / 197 Amber_Under / 1 Amber_Tight / 0 Red. Most routes show as Amber_Under because the engine sizes fleet for the target headway irrespective of current demand — by design, the plan delivers more seats than today's ridership uses, to enable the latent demand that frequent service unlocks. The Mohring effect (riders shift to bus when service improves) is not modelled — that is a known v4 backlog item.

5. How the engine decides which headway to give which route

Headway is assigned per **Priority Band** (HP / MP / LP), and the band itself is computed from a **Composite Demand Index**:

```
Final_CDI = Pop_Score × 0.5 + POI_Score × 0.5  
Pop_Score – normalised population in 400 m walk catchment  
POI_Score – gravity score of weighted POIs in 250 m buffer (3-tier: year-round 1.0, secondary 0.4, tourist 0.6/0.0)
```

The 207 active routes are then split into HP / MP / LP using **Jenks Natural Breaks** on the CDI distribution. Routes near a band boundary are nudged up or down by a **Road_Multiplier** tie-breaker (highway/arterial = 1.0, mohalla lane = 0.85). The SSCL backbone bypasses all of this and is forced to HP.

Overrides — when band logic doesn't apply

- **SSCL backbone** — 45 permits matched to the 30 official CHALO routes are forced HP / 15-min headway.
- **Social Obligation floor** — routes serving KP townships, district hospitals, and women's colleges get a LP→MP promotion (11 anchor keys after the v3.3.1 audit).
- **District HQ floor** — Regional_District routes that touch a district headquarters also get LP→MP.
- **Tourist corridor boost** — routes within 2 km of a distant tourist destination (Gulmarg, Pahalgam, Sonamarg, etc.) or with an endpoint within 0.6 km of an inner-city tourist gate (Mughal gardens, Boulevard) get a 1.3× catchment-population boost, which raises Pop_Score → CDI → band.

6. Sanity-checking against CHALO ground truth

The Kashmir engine is calibrated against twelve months of CHALO ridership data (May 2025 – April 2026, 11.6 M trips across the 30 SSCL e-bus routes). The calibration target is per-route fleet at the same target headway:

CHALO scaled fleet = CHALO_current_fleet ×
(CHALO_effective_headway ÷ engine_target_headway) = 98 × (34 min ÷
15 min) ≈ 220 buses ⇒ ~7.33 buses per route at 15-min headway

v3.3.5 calibration scorecard

Metric	CHALO (current Op)	CHALO (scaled to 15-min)	Engine v3.3.5	Error
SSCL fleet (raw count)	98	~220	362	+64% (cycle-time conservatism)
SSCL fleet per route	3.27	7.33	8.04	+9.7% (within ±25%)
SSCL Daily_Demand_Pax	31,869	n/a	~33,000	+3.4% (within ±10%)
Tourist corridors flagged	n/a	n/a	69	✓
Red_Overload routes	n/a	n/a	0	✓
QC checks	n/a	n/a	8/8	✓

The 'scaled CHALO' column is the apples-to-apples comparator: if CHALO themselves ran at the engine's 15-min target headway (instead of their current ~34-min effective headway), they'd need ~220 buses instead of 98. Engine recommends 362, of which the extra 142 is largely the absorption of duplicate private/JKRTC permits running parallel to SSCL e-bus corridors (currently the SSCL e-bus shares the road with these operators; the plan upgrades them to trunk service alongside SSCL).

7. Peer-city benchmarks — is the plan realistic?

Public-transit research uses **buses per 1,000 residents** as the cross-city scale-invariant comparator. Indian peer cities:

City	Population (urban)	Bus fleet	Buses / 1000 res.
Mysuru (KSRTC)	2.0 M	~700	0.35
Bhopal (BCLL)	1.4 M	~600	0.43
Indore (Atal Indore)	2.0 M	~900	0.45
Bengaluru (BMTc)	12.8 M	~6,500	0.51
Srinagar v3.3.5	1.66 M	988	0.60
Chandigarh (CTU)	1.0 M	~650	0.65
Pune (PMPML)	3.7 M	~2,800	0.75
Delhi (DTC + cluster)	7.8 M	~7,000	0.90
Mumbai (BEST)	2.9 M	~3,500	1.20

Srinagar at 0.60 buses per 1,000 residents sits squarely between BMTc Bengaluru (0.51) and Chandigarh CTU (0.65 — closest peer city by size). The v3.3.4 'aspirational' plan at 0.67 buses/1000 — driven by 15-min headway on all 130 HP routes — was an achievable Year-3+ ambition; v3.3.5 is the defensible Year-1 plan.

Vehicle-density vs service-quality trade-off

More buses do not automatically buy better service. The right metric is the joint distribution of (a) headway on the busiest corridors, and (b) coverage breadth — what fraction of residents live within 400 m of some bus stop. The v3.3.5 plan achieves both: 130 trunks at 15–20 min headway, plus 77 feeders covering 69.78% of the Srinagar UA + Valley population.

Appendix A — Glossary of engine terms

Term	Meaning
Headway	Time between consecutive buses on the same route.
Cycle time	Wall-clock time for one bus to complete one full round trip.
CDI	Composite Demand Index = $\text{Pop_Score} \times 0.5 + \text{POI_Score} \times 0.5$
HP / MP / LP	Jenks-band classification: high / medium / low priority.
SSCL	Srinagar Smart City Limited — operator of the 30-route e-bus backbone.
CHALO	Free-fare ridership-tracking system on SSCL buses; 12-month dataset is the engine's calibration anchor.
CMP_Trunk	Boolean flag: route is one of the 45 matched SSCL permits forced to trunk service.
Action_Taken	Per-route disposition: RETAINED_AS_FEEDER / UPGRADED_TO_TRUNK / MERGED_INTO_TRUNK.
Load_Ratio	$\text{Daily_Demand_Pax} \div \text{Daily_Capacity_Pax}$ — Phase-4 diagnostic.
Spare ratio	Industry-standard 1.15 multiplier on operating fleet to cover maintenance, breakdowns, depot rotation.
Mode share	Fraction of resident trips taken by bus. 9% urban / 7.2% peri-urban / 5.4% regional (typology-aware, CHALO)
Trip rate	Average bus trips per resident per day. 1.6, weighted across female (~2.0) and male (~1.2) segments.
CAPTURE_SCALE	0.18 — empirical scalar tying buffer-based demand to CHALO observed ridership.
Jenks breaks	Optimal natural-break clustering algorithm for splitting a 1D distribution into bands.

Appendix B — Engine constants and where they come from

Constant	Value	Source
HEADWAY_HP_MIN (non-SSCL)	20 min	Phase-1 peer-city target (v3.3.5)
HEADWAY_MP_MIN	35 min	Peer-city feeder norm (v3.3.5)
HEADWAY_LP_MIN	60 min	Inter-district floor
SSCL_TRUNK_HEADWAY_MIN	15 min	SSCL published design target
FLEET_SPARE_RATIO	1.15	Industry-standard 10–20% spare
MIN_FLEET_URBAN	2	Reliability floor on urban routes
MIN_FLEET_REGIONAL	1	Lifeline-route floor (v3.2 audit)
CITY_CORE_LAT_THRESHOLD	34.07°	Srinagar Lal Chowk + Downtown latitude
CONGESTION_CITY_CORE	2.2×	Srinagar Traffic Police peak studies
CONGESTION_PERI_URBAN	1.4×	Same source
JHELMUM_BRIDGE_BOTTLENECK_MIN	8 min	Habba Kadal / Amira Kadal observed queue
STOP_PENALTY_MIN	0.5 min	30-second standard dwell
WALK_CATCHMENT_M	400 m	5-minute walk; TRB transit-coverage standard
WINTER_WALKSHED_SHRINK	0.65	Chillai Kalan snow conditions (35% shrink)
VEHICLE_CAPACITY_HP_V	50 seats	12-metre standard low-floor
VEHICLE_CAPACITY_MP_V	35 seats	9-metre SSCL e-bus / midi
VEHICLE_CAPACITY_LP_V	15 seats	Tempo / Magic / small minibus
TOURIST_POPULATION_MULTIPLIER	1.3×	Tier-3 POI catchment uplift (v3.3.3)
PHASE4_MODE_SHARE (urban)	0.09	CHALO 12-month observed bus share
PHASE4_TRIP_RATE	1.6	Weighted female (2.0) + male (1.2) avg
PHASE4_CORRIDOR_CAPTURE_SCALE	0.18	Empirical SSCL anchor (v3.3.2)

Appendix C — Where each concept lives in the code

Concept	File / function
Headway assignment	transit_kashmir_v3.py → step6_assign_headways()
Cycle time	transit_kashmir_v3.py → compute_cycle_times()
Fleet sizing	transit_kashmir_v3.py → step8_compute_fleet()
SSCL empirical floor	transit_kashmir_v3.py → step9_assign_vehicle_split() (v3.3.4)
Daily_Capacity (corrected)	transit_kashmir_v3.py → compute_phase4_metrics() (v3.3.3 T3)
Daily_Demand corridor share	transit_kashmir_v3.py → compute_phase4_metrics() (v3.3.2)
Tourist tagging (geometric)	transit_kashmir_v3.py → _line_near_point_km() (v3.3.3 T1)
CDI / Jenks bands	transit_kashmir_v3.py → step4a_compute_final_cdi() + step5_assign_priority_bands()
CHALO cross-validation	cross_evaluate.py → evaluate()
Briefing-deck generator	generate_presentations.py
Route-code generator	generate_route_codes.py (needs Kashmir_Stops_Sector_V2.csv)

Reading order for someone new to the engine

1. README.md (project overview, audit lineage v3.1 → v3.3.5)
2. Sections 1–3 of this guide (headway, cycle time, fleet formula)
3. Section 4 (capacity vs demand) and section 6 (CHALO calibration)
4. Engine source — top-of-file constants block lines 130–500
5. Run the engine once with the real OSRM Docker up; inspect outputs_v3.3.5/Rationalised_Routes_Kashmir_v3.csv against the dashboard.
6. Run cross_evaluate.py to see how every number ties back to CHALO observations.